

Causal Inference

Lecture 4

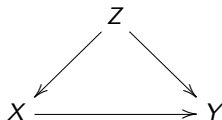
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- 2 Standardization
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- Simple causal structure where Z confounds the effect of X on Y .
- Z is the only confounder.
- Our goal is to identify and estimate X to Y .

- In the observed data:

$$Y = \begin{cases} Y_1 & \text{if } X = 1, \\ Y_0 & \text{if } X = 0. \end{cases}$$

- The causal effect of X on Y can be defined as:

$$\text{Individual Causal Effect} = g(Y_1, Y_0).$$

where g is some contrast.

- Usually we are interest in the causal effect on the population level. For example, the average (causal) treatment effect (ATE)

$$\text{ATE} = \mathbb{E}[Y_1 - Y_0].$$

- The observed outcome Y depends on the actual exposure X :

$$Y = XY_1 + (1 - X)Y_0.$$

- The definition above illustrates the consistency assumption.

The Fundamental Problem of Causal Inference

Definition (Holland, 1986)

The Fundamental Problem of Causal Inference states:

For any individual, we can observe only one of the counterfactual outcomes: Y_1 or Y_0 .

- Y_1 : Outcome if the individual receives the treatment ($X = 1$).
- Y_0 : Outcome if the individual does not receive the treatment ($X = 0$).
- Problem:

Y_1 and Y_0 cannot be observed simultaneously for the same individual.

Implications

- We cannot directly compute the causal effect for an individual, i.e., the $Y_1 - Y_0$.
- Causal inference relies on comparing outcomes across individuals or groups based on appropriate assumptions (e.g., randomization or statistical models).

- Assume a parametric model for Y :

$$Y = f(X, Z; \beta) + \epsilon,$$

where:

- $f(X, Z; \beta)$: Specifies the relationship between X , Z , and Y .
 - β : Parameters of the model.
 - ϵ : Error term.
- The counterfactual outcome Y_x is:

$$Y_x = f(x, Z; \beta) + \epsilon.$$

- Example (linear model):

$$Y_x = \beta_0 + \beta_1 x + \beta_2 Z + \epsilon.$$

- This average treatment effect (ATE) is defined as

$$\text{ATE} = \mathbb{E}[Y_1 - Y_0].$$

- For a binary outcome it equals the causal risk difference

$$\text{ATE} = \mathbb{P}(Y_1 = 1) - \mathbb{P}(Y_0 = 1).$$

- Under the linear parametric model:

$$\text{ATE} = \mathbb{E}[Y_1 - Y_0] = (\beta_0 + \beta_1 1 + \beta_2 Z) - (\beta_0 + \beta_1 0 + \beta_2 Z) = \beta_1,$$

where β_1 is the coefficient of X in:

$$\mathbb{E}[Y|X, Z] = \beta_0 + \beta_1 X + \beta_2 Z.$$

- Assumptions:
 - $\mathbb{E}[\epsilon | X, Z] = 0$: No unmeasured confounding.
 - Model $\mathbb{E}[Y|X, Z]$ is correctly specified.

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- If we have conditional exchangeability, given Z , then

$$\begin{aligned}\mathbb{P}(Y_0 = 1) &= \sum_Z \mathbb{P}(Y = 1|X = 0, Z)\mathbb{P}(Z) \\ \mathbb{P}(Y_1 = 1) &= \sum_Z \mathbb{P}(Y = 1|X = 1, Z)\mathbb{P}(Z)\end{aligned}$$

- If Z is binary, then we can estimate $\mathbb{P}(Y = 1|X = 0, Z)$ and $\mathbb{P}(Y = 1|X = 1, Z)$ without modeling assumptions
- If Z is continuous, this non-parametric approach is not feasible
 - very few subjects for each observed level of Z
- But with a regression model we can 'extrapolate', to estimate both $\mathbb{P}(Y = 1|X = 0, Z)$ and $\mathbb{P}(Y = 1|X = 1, Z)$

Four steps for standardization with regression model

The aim is to estimate the average treatment effect (ATE). This goal is achieved by standardization (or adjusting to Z), i.e.,

$$\begin{aligned}\mathbb{E}[Y_1 - Y_0] &= \mathbb{P}(Y_1 = 1) - \mathbb{P}(Y_0 = 1) \\ &= \mathbb{E}[\mathbb{E}[Y_1 - Y_0|Z]] \\ &= \mathbb{E}[\mathbb{E}[Y_1|X = 1, Z] - \mathbb{E}[Y_0|X = 0, Z]] \\ &= \mathbb{E}_Z[\mathbb{E}[Y|X = 1, Z] - \mathbb{E}[Y|X = 0, Z]] \\ &= \mathbb{E}_Z[\mathbb{P}(Y = 1|X = 1, Z) - \mathbb{P}(Y = 1|X = 0, Z)]\end{aligned}$$

- **Step 1:** fit a regression model for the outcome
- **Step 2:** replace the factual level of X with 0 for each subject
- **Step 3:** use the fitted model to estimate $\mathbb{P}(Y = 1|X, Z)$ for each subject, i.e. for each level of (X, Z)
- **Step 4:** average these estimates to obtain an estimate of $\mathbb{P}(Y_0 = 1)$
- To estimate $\mathbb{P}(Y_1 = 1)$, replace X with 1 in step 2

Example

Using the aids data: Y is the binary outcome, X is the binary exposure, Z is the continuous confounder.

1. Assume that outcome logistic model

$$\mathbb{E}[Y|X, Z] = \mathbb{P}(Y = 1|X, Z) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ)}}$$

2. Use the estimated parameters β^T , and for every $i = 1, \dots, n$, fit the outcome for $X = 0$ and $X = 1$, i.e.,

$$\hat{\mathbb{P}}(Y = 1|X = 0, Z = z_i) = \frac{1}{1 + e^{-(\hat{\beta}_0 + \hat{\beta}_1 0 + \hat{\beta}_2 z_i + \beta_3 0 z_i)}}$$

$$\hat{\mathbb{P}}(Y = 1|X = 1, Z = z_i) = \frac{1}{1 + e^{-(\hat{\beta}_0 + \hat{\beta}_1 1 + \hat{\beta}_2 z_i + \beta_3 1 z_i)}}$$

3. Estimate $\mathbb{P}(Y_0 = 1)$ by the sample average of $\hat{\mathbb{P}}(Y = 1|X = 0, Z = z_i)$, and $\mathbb{P}(Y_1 = 1)$ by the sample average of $\hat{\mathbb{P}}(Y = 1|X = 1, Z = z_i)$.
4. Hence, the standardized estimator of ATE (which is the risk difference: $\mathbb{P}(Y_1 = 1) - \mathbb{P}(Y_0 = 1)$ for a binary outcome) is

$$\begin{aligned}\widehat{ATE}_{stn} &= \hat{\mathbb{P}}(Y_1 = 1) - \hat{\mathbb{P}}(Y_0 = 1) \\ &= \frac{1}{n} \sum_{i=1}^n \hat{\mathbb{P}}(Y = 1|X = 1, Z = z_i) - \frac{1}{n} \sum_{i=1}^n \hat{\mathbb{P}}(Y = 1|X = 0, Z = z_i) \\ &= \frac{1}{n} \sum_{i=1}^n \frac{1}{1 + e^{-(\hat{\beta}_0 + \hat{\beta}_1 1 + \hat{\beta}_2 z_i + \beta_3 1 z_i)}} - \frac{1}{n} \sum_{i=1}^n \frac{1}{1 + e^{-(\hat{\beta}_0 + \hat{\beta}_1 0 + \hat{\beta}_2 z_i + \beta_3 0 z_i)}}.\end{aligned}$$

- This is also known as the G-computation formula (Robins, 1986).

Code for standardization with regression model

```
# Step 1: Fit the logistic regression model
model = sm.Logit(y, X)
fit = model.fit()

# Step 2: Create a copy of the dataset for X=0
aids0 = aids.copy()
aids0["X"] = 0 # Set X to 0
aids0["X:Z"] = aids0["X"] * aids0["Z"] # Update interaction term

# Step 3: Predict probabilities for X=0
X0 = aids0[["X", "Z", "X:Z"]]
X0 = sm.add_constant(X0) # Add intercept
pred0 = fit.predict(X0)

# Step 4: Calculate the mean predicted probability for X=0
p0 = pred0.mean()
print("Mean predicted probability for X=0:", p0)
Mean predicted probability for X=0: 0.6090828480763144
```

```
# Step 1: Fit the logistic regression model
model = sm.Logit(y, X)
fit = model.fit()

# Step 2: Create a copy of the dataset for X=1
aids1 = aids.copy()
aids1["X"] = 1 # Set X to 1
aids1["const"] = 1

# Step 3: Predict probabilities for X=1
X1 = aids1[["const", "X", "Z"]]
X1["X:Z"] = aids1["X"] * aids1["Z"]

pred1 = fit.predict(X1)

# Step 4: Calculate the mean predicted probability for X=1
p1 = pred1.mean()
print("Mean predicted probability for X=1:", p1)
Mean predicted probability for X=1: 0.06333355735321514
```

The marginal causal log odds ratio

$$\begin{aligned}\hat{\mathbb{P}}(Y_0 = 1) &= 0.6090828 \\ \hat{\mathbb{P}}(Y_1 = 1) &= 0.06333356\end{aligned}$$

- We can use $\hat{\mathbb{P}}(Y_0 = 1)$ and $\hat{\mathbb{P}}(Y_1 = 1)$ to estimate of the marginal causal log odds ratio

$$\log \left\{ \frac{\hat{\mathbb{P}}(Y_1 = 1)}{1 - \hat{\mathbb{P}}(Y_1 = 1)} / \frac{\hat{\mathbb{P}}(Y_0 = 1)}{1 - \hat{\mathbb{P}}(Y_0 = 1)} \right\} = -3.14$$

$$\begin{aligned}\hat{\mathbb{P}}(Y_0 = 1) &= 0.6090828 \\ \hat{\mathbb{P}}(Y_1 = 1) &= 0.06333356\end{aligned}$$

- Once we have estimated $\hat{\mathbb{P}}(Y_1 = 1)$ and $\hat{\mathbb{P}}(Y_0 = 1)$ we can estimate any measure of marginal causal effect, e.g.

$$\begin{aligned}\text{causal risk difference} &= \hat{\mathbb{P}}(Y_1 = 1) - \hat{\mathbb{P}}(Y_0 = 1) = -0.55 \\ \text{causal risk ratio} &= \hat{\mathbb{P}}(Y_1 = 1) / \hat{\mathbb{P}}(Y_0 = 1) = 0.10\end{aligned}$$

even though the estimates were derived from a logistic regression model

The package stdReg

```
> library(stdReg)
> fit <- glm(formula=Y~X+Z+X*Z, family=binomial,
data=aids)
> std.fit <- stdGlm(fit=fit, data=aids, X="X")
> summary(std.fit)
```

Formula: $Y \sim X + Z + X * Z$

Family: binomial

Link function: logit

Exposure: X

	Estimate	Std. Error	lower 0.95	upper 0.95
0	0.6091	0.00655	0.5962	0.6219
1	0.0633	0.00348	0.0565	0.0702

```
> summary(object=std.fit, transform="logit",  
contrast="difference", reference=0)
```

Formula: $Y \sim X + Z + X * Z$

Family: binomial

Link function: logit

Exposure: X

Transform: logit

Reference level: $X = 0$

Contrast: difference

	Estimate	Std. Error	lower 0.95	upper 0.95
0	0.00	0.0000	0.00	0.00
1	-3.14	0.0644	-3.26	-3.01

```
# Load the dataset
aids = pd.read_csv("aids.txt", sep=" ")

# Compute interaction term manually
aids["X:Z"] = aids["X"] * aids["Z"]

# Create the design matrix (manual intercept)
X = aids[["X", "Z", "X:Z"]]
X.insert(0, "intercept", 1)
y = aids["Y"]

# Fit the logistic regression model
model = sm.Logit(y, X)
fit = model.fit()

# Predict log-odds for X=0
aids0 = aids.copy()
aids0["X"] = 0
aids0["X:Z"] = aids0["X"] * aids0["Z"]
X0 = aids0[["X", "Z", "X:Z"]]
X0.insert(0, "intercept", 1)
log_odds0 = fit.predict(X0, linear=True)

# Predict log-odds for X=1
aids1 = aids.copy()
aids1["X"] = 1
aids1["X:Z"] = aids1["X"] * aids1["Z"]
X1 = aids1[["X", "Z", "X:Z"]]
X1.insert(0, "intercept", 1)
log_odds1 = fit.predict(X1, linear=True)

# Calculate the average difference
difference = log_odds1.mean() - log_odds0.mean()
print(f"Difference in log-odds (X=1 vs X=0): {difference:.4f}")

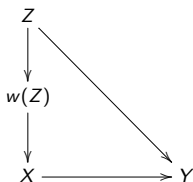
Difference in log-odds (X=1 vs X=0): -3.3633
```

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- Assume that Z is sufficient for confounding control, namely,

$$Y_1, Y_0 \perp\!\!\!\perp X|Z.$$

- Although Z can be high-dimensional, in order to get the conditional exchangeability, it is suffice to condition on a scalar $\mathbb{P}(X = 1|Z) = w(Z)$, which is known as the propensity score.
- Graphically,



namely, conditioning on the propensity score $z(W)$ blocks the path from Z to X ,

$$Y_1, Y_0 \perp\!\!\!\perp X|w(Z).$$

- The idea is to create pseudo-population by reweighing the observed data using the propensity score.
- In the pseudo-population the exposure X and the confounder Z are independent, namely,

$$\mathbb{P}(X = x|Z) = \mathbb{P}(X = x).$$

- Individuals with $X = 1$ are reweighted with $1/w(Z)$, and individuals with $X = 0$ are reweighted with $1/(1 - w(Z))$ to create a pseudo-population where the treatment assignment is independent of the confounders.
- For the binary treatment X the ATE is

$$\begin{aligned}\text{ATE} &= \mathbb{E}[Y_1 - Y_0] \\ &= \mathbb{E}[Y_1] - \mathbb{E}[Y_0] \\ &= \mathbb{E}\left[\frac{XY}{w(Z)}\right] - \mathbb{E}\left[\frac{(1-X)Y}{1-w(Z)}\right].\end{aligned}$$

- Assumptions for the estimator to hold:
 1. Conditional exchangeability $Y_1, Y_0 \perp\!\!\!\perp X|Z$.
 2. Positivity: $w(Z) \in (0, 1)$.

Proof of the consistency of the IPW estimator

For the first term

$$\begin{aligned}\mathbb{E}\left[\frac{XY}{w(Z)}\right] &= \mathbb{E}\left[\frac{X}{w(Z)}(Y_1X + (1-X)Y_0)\right] \\&= \mathbb{E}\left[\frac{X}{w(Z)}XY_1 + \frac{X}{w(Z)}(1-X)Y_0\right] \text{ where } X^2Y_1 = XY_1, X(1-X) \equiv 0 \\&= \mathbb{E}_Z\left[\mathbb{E}\left[\frac{X}{w(Z)}Y_1\right] | Z\right] \text{ iterated expectation} \\&= \mathbb{E}_Z\left[\frac{1}{w(Z)}\mathbb{E}[XY_1|Z]\right] \\&= \mathbb{E}_Z\left[\frac{1}{w(Z)}\mathbb{E}[X|Z]\mathbb{E}[Y_1|Z]\right] \quad Y_1 \perp\!\!\!\perp X|Z \\&= \mathbb{E}_Z\left[\frac{1}{w(Z)}w(Z)\mathbb{E}[Y_1|Z]\right] \\&= \mathbb{E}_Z[\mathbb{E}[Y_1|Z]] \\&= \mathbb{E}[Y_1] \quad \square\end{aligned}$$

The proof for the second term follows along the same lines.

- The sample version is straightforward

$$\widehat{ATE}_{IPW} = \sum_{i=1}^n \frac{x_i y_i}{\hat{w}(z_i)} - \sum_{i=1}^n \frac{(1 - x_i) y_i}{1 - \hat{w}(z_i)}.$$

- Alternatively, the IPW estimator can be written in the following form (known as the Horvitz and Thompson (1952) estimator)

$$\begin{aligned} \widehat{ATE}_{IPW} &= \hat{\mathbb{P}}(Y_1 = 1) - \hat{\mathbb{P}}(Y_1 = 0) \\ &= \frac{\sum_{i: x_i=1} y_i / \hat{w}(z_i)}{\sum_{i: x_i=1} 1 / \hat{w}(z_i)} - \frac{\sum_{i: x_i=0} y_i / (1 - \hat{w}(z_i))}{\sum_{i: x_i=0} 1 / (1 - \hat{w}(z_i))}, \end{aligned}$$

where $\sum_{i: x_i=1} 1 / \hat{w}(z_i)$ is the number of exposed individuals in the reweighted pseudo population, and $\sum_{i: x_i=0} 1 / (1 - \hat{w}(z_i))$ is the number of unexposed individuals in the reweighted pseudo population.

Three steps for IPW with regression model

- **Step 1:** fit a regression model for the exposure
- **Step 2:** use the fitted model to estimate the subject-specific weight (a.k.a. propensity score)

$$W = \frac{1}{p(X|Z)}$$

- **Step 3:** use $p(Y = 1|X = 0)$ and $p(Y = 1|X = 1)$ in the weighted sample as estimates of $p(Y_0 = 1)$ and $p(Y_1 = 1)$

Example

Using the aids data: Y is the binary outcome, X is the binary exposure, Z is the continuous confounder.

1. Assume that the exposure follows a logistic model

$$\mathbb{E}[X|Z] = \mathbb{P}(X = 1|Z) = w(Z) = \frac{1}{1 + e^{-(\alpha_0 + \alpha_1 Z)}}$$

2. For every individual we use his/her z_i and the estimate α^T to estimate $w(z_i)$
3. The IPW estimator of ATE (which is the risk difference: $\mathbb{P}(Y_1 = 1) - \mathbb{P}(Y_0 = 1)$ for a binary outcome) is

$$\widehat{\text{ATE}}_{IPW} = \frac{1}{n_1} \sum_{i:x_i=1} \frac{y_i}{\hat{w}(z_i)} - \frac{1}{n_0} \sum_{i:x_i=0} \frac{y_i}{1 - \hat{w}(z_i)},$$

where $n_1 = \sum_{i:x_i=1} \frac{1}{\hat{w}(z_i)}$, and $n_0 = \sum_{i:x_i=0} \frac{1}{1 - \hat{w}(z_i)}$.

Code for IPW with regression model in R

```
> #step 1
> fit <- glm(formula=X~Z, family=binomial,
data=aids)
> #step 2
> pred <- predict(object=fit, type="respons")
> w <- 1/(aids$X*pred+(1-aids$X)*(1-pred))
> #step 3 for x=0
> p0 <- weighted.mean(aids$Y[aids$X==0],
w[aids$X==0])
> p0
[1] 0.6086676
> #step 3 for x=1
> p1 <- weighted.mean(aids$Y[aids$X==1],
w[aids$X==1])
> p1
[1] 0.06261394
```

Code for IPW with regression model in Python

```
# Step 1: Load the saved DataFrame
aids_df = pd.read_csv("aids.txt", sep=" ")

# Step 2: Fit logistic regression
X = add_constant(aids_df['Z']) # Add intercept
y = aids_df['X']
logit_model = Logit(y, X).fit()

# Predict probabilities
aids_df['pred'] = logit_model.predict(X)

# Calculate weights
aids_df['w'] = 1 / (aids_df['X'] * aids_df['pred'] +
                   (1 - aids_df['X']) * (1 - aids_df['pred']))

# Step 3: Calculate weighted mean for X = 0
p0 = np.average(aids_df.loc[aids_df['X'] == 0, 'Y'],
               weights=aids_df.loc[aids_df['X'] == 0, 'w'])

# Step 3: Calculate weighted mean for X = 1
p1 = np.average(aids_df.loc[aids_df['X'] == 1, 'Y'],
               weights=aids_df.loc[aids_df['X'] == 1, 'w'])

p0, p1
(0.6086675565580764, 0.06261393809533125)
```

The marginal causal log odds ratio

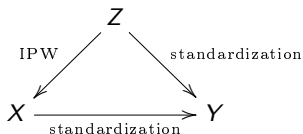
$$\begin{aligned}\hat{\mathbb{P}}(Y_0 = 1) &= 0.6086676 \\ \hat{\mathbb{P}}(Y_1 = 1) &= 0.06261394\end{aligned}$$

- We can use $\hat{\mathbb{P}}(Y_0 = 1)$ and $\hat{\mathbb{P}}(Y_1 = 1)$ to estimate of the marginal causal log odds ratio

$$\log \left\{ \frac{\hat{\mathbb{P}}(Y_1 = 1)}{1 - \hat{\mathbb{P}}(Y_1 = 1)} / \frac{\hat{\mathbb{P}}(Y_0 = 1)}{1 - \hat{\mathbb{P}}(Y_0 = 1)} \right\} = -3.15$$

... but we can also use $\hat{\mathbb{P}}(Y_0 = 1)$ and $\hat{\mathbb{P}}(Y_1 = 1)$ to estimate any other measure of marginal causal effect

- e.g. the causal risk difference of risk ratio



- Standardization and IPW require different models
 - standardization requires an outcome model, describing how the outcome depends on the exposure and confounders
 - IPW requires an exposure model, describing how the exposure depends on the confounders
- If we know more about how the confounders affect the exposure, then more natural to use an exposure model
- If we know more about how the confounders affect the outcome, then more natural to use an outcome model

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- The doubly robust (DR) estimator combines two approaches:
 - Outcome Model: Models the outcome Y based on covariates Z and treatment X .
 - Propensity Score Model: Models the probability of treatment assignment given Z , $w(Z) = \mathbb{P}(X = 1 | Z)$.
- Key property: The DR estimator is consistent if either:
 - The outcome model is correctly specified, or
 - The propensity score model is correctly specified.
- This provides robustness to model misspecification.

- The doubly robust estimator for the ATE is:

$$\text{ATE}_{\text{DR}} = \mathbb{E} \left[\frac{X \cdot (Y - \mu_1(Z))}{w(Z)} + \mu_1(Z) \right] - \mathbb{E} \left[\frac{(1 - X) \cdot (Y - \mu_0(Z))}{1 - w(Z)} + \mu_0(Z) \right],$$

where:

- $\mu_1(Z) = \mathbb{E}[Y \mid X = 1, Z]$ is the predicted outcome for $X = 1$,
 - $\mu_0(Z) = \mathbb{E}[Y \mid X = 0, Z]$ is the predicted outcome for $X = 0$,
 - $w(Z) = \mathbb{P}(X = 1 \mid Z)$ is the propensity score.
- This combines:
 - Outcome regression: $\mu_1(Z), \mu_0(Z)$,
 - Inverse probability weighting: $1/w(Z), 1/(1 - w(Z))$.

Proof of the consistency of the Doubly robust estimator

For the first term

$$\begin{aligned}\mathbb{E} \left[\frac{X(Y - \mu_1(Z))}{w(Z)} + \mu_1(Z) \right] &= \mathbb{E} \left[\frac{X(XY_1 + (1 - X)Y_0 - \mu_1(Z))}{w(Z)} + \mu_1(Z) \right] \text{ consistency} \\ &= \mathbb{E} \left[\frac{X(Y_1 - \mu_1(Z))}{w(Z)} + \mu_1(Z) \right] \\ &= \mathbb{E}_! \left[\mathbb{E} \left[\frac{X(Y_1 - \mu_1(Z))}{w(Z)} + \mu_1(Z) \right] | Z \right] \text{ iterated expectation} \\ &= \mathbb{E}_Z \left[\frac{\mathbb{E}[X|Z](\mathbb{E}[Y_1|Z] - \mu_1(Z))}{w(Z)} + \mu_1(Z) \right] Y_1 \perp\!\!\!\perp X|Z\end{aligned}$$

- If the exposure model is correctly specified, then $\mathbb{E}[X|Z] = w(Z)$ and the expression boils down to $\mathbb{E}_Z[\mathbb{E}[Y_1|Z]] = \mathbb{E}[Y_1]$.
- If the outcome model is correctly specified, then $\mathbb{E}[Y_1|Z] = \mathbb{E}[Y_1|Z, X = 1] = \mathbb{E}[Y|Z, X = 1] = \mu_1(Z)$ and the expression boils down to $\mathbb{E}_Z[\mu_1(Z)] = \mathbb{E}_Z[\mathbb{E}[Y|Z, X = 1]] = \mathbb{E}_Z[\mathbb{E}[Y_1|Z, X = 1]] = \mathbb{E}_Z[\mathbb{E}[Y_1|Z]] = \mathbb{E}[Y_1]$.
- The proof for the second term follows along the same lines.

1. Fit the propensity score model:
 - Estimate $w(Z) = \mathbb{P}(X = 1 \mid Z)$ using logistic regression.
2. Fit the outcome model:
 - Estimate $\mu_1(Z) = \mathbb{E}[Y \mid X = 1, Z]$,
 - Estimate $\mu_0(Z) = \mathbb{E}[Y \mid X = 0, Z]$.
3. Combine both models:

$$\widehat{\text{ATE}}_{\text{DR}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{X_i(Y_i - \hat{\mu}_1(Z_i))}{\hat{w}(Z_i)} + \hat{\mu}_1(Z_i) \right] - \frac{1}{n} \sum_{i=1}^n \left[\frac{(1 - X_i)(Y_i - \hat{\mu}_0(Z_i))}{1 - \hat{w}(Z_i)} + \hat{\mu}_0(Z_i) \right].$$

Example: Doubly Robust Estimation with aids data - in R

```
aids = read.csv("aids.txt", sep = " ")

# 1. estimate the exposure model
exp_m <- glm( X ~ Z, data = aids, family = "binomial")

# estimate the weights
w_Z <- function(Z) { plogis(coef(exp_m)[1] + coef(exp_m)[2] * Z) }

# 2. Estimate the outcome model
out <- glm( Y ~ X + Z + X * Z, data = aids, family = "binomial")

# fit the outcome function
out_X_Z <- function(X, Z) { plogis(coef(out)[1] + coef(out)[2] * X +
  coef(out)[3] * Z +
  coef(out)[4] * X * Z) }

# 3. estimate the DR estimator of the risk-difference
mean( aids$X * (aids$Y - out_X_Z(X = 1, Z = aids$Z)) / w_Z(Z = aids$Z)
  + out_X_Z(X = 1, Z = aids$Z) ) -
mean( (1 - aids$X) * (aids$Y - out_X_Z(X = 0, Z = aids$Z)) / (1 - w_Z(Z = aids$Z))
  + out_X_Z(X = 0, Z = aids$Z) )
```

Example: Doubly Robust Estimation with aids data - in Python

```
import pandas as pd
import numpy as np
from statsmodels.api import Logit, add_constant
from scipy.special import expit

# Load the dataset
aids = pd.read_csv("aids.txt", sep=" ")

# 1. Estimate the exposure model
# Logistic regression for exposure ( $X \sim Z$ )
X_exposure = add_constant(aids["Z"]) # Add intercept
y_exposure = aids["X"]
exp_m = Logit(y_exposure, X_exposure).fit()

# Function to calculate propensity scores ( $w_Z$ )
def w_Z(Z):
    return expit(exp_m.params[0] + exp_m.params[1] * Z)

# 2. Estimate the outcome model
# Logistic regression for the outcome ( $Y \sim X + Z + X * Z$ )
aids["X_Z"] = aids["X"] * aids["Z"] # Interaction term
X_outcome = add_constant(aids[["X", "Z", "X_Z"]]) # Add intercept and terms
y_outcome = aids["Y"]
out = Logit(y_outcome, X_outcome).fit()
```

Example: Doubly Robust Estimation with aids data - in Python (continued)

```
# Function to calculate predicted outcomes (out_X_Z)
def out_X_Z(X, Z):
    # Ensure 'const' is included in the data frame for prediction
    out1 = expit(out.params[0] + out.params[1] * X + out.params[2] * Z
                + out.params[3] * X * Z)

    return out1

# 3. Estimate the DR estimator of the risk-difference
# Calculate the doubly robust estimator
dr_rate = ( np.mean(aids["X"] * (aids["Y"] - out_X_Z(1, aids["Z"]))) /
            w_Z(aids["Z"]))
+ out_X_Z(1, aids["Z"]))
-
np.mean(((1 - aids["X"]) * (aids["Y"] - out_X_Z(0, aids["Z"]))) /
        (1 - w_Z(aids["Z"])))
+ out_X_Z(0, aids["Z"])))
```

- The variance of the doubly robust estimator is not straightforward to compute analytically.
- Although there are some approximations, the variance can be challenging to derive due to the interaction between the outcome regression and propensity score models.
- At this stage, we can use bootstrap methods to compute the estimator's sample variance and confidence intervals.
- For large samples, the variance can be computed using the sandwich formula.
- In some situations, the Delta method can also be applied when the estimator involves differentiable functions of estimated parameters.
- We will revisit this topic in future classes, particularly in the context of G-estimation.

- 1 Causal effects and models
- 2 Standardization
- 3 Inverse probability weighting
- 4 Doubly robust estimator**
 - Bootstrap
- 5 Summary

Definition

The bootstrap (BS) is a numeric method for estimating standard errors and computing confidence intervals. The underlying idea is to treat the sample as the population.

Algorithm

Let $X_1, \dots, X_n$ be a random sample from $F_X(x; \theta)$, with empirical distribution $\hat{F}_n$. Let the parameter of interest be $g(\theta)$, thus

1. Draw $X_1^*, \dots, X_n^* \sim \hat{F}_n$.
2. Compute $g_b^*(\hat{\theta}(X_1^*, \dots, X_n^*))$.
3. Repeat steps 1 and 2, B times, to get $g_1^*, \dots, g_B^*$.
4. Use the empirical quantiles of $g_1^*, \dots, g_B^*$ to construct the BS empirical percentile-based CIs or estimate sample variance of $g(\theta)$'s estimator with

$$\frac{1}{B} \sum_{b=1}^B \left(g_b^* - \frac{1}{B} \sum_{b=1}^B g_b^* \right)^2.$$

For the Doubly Robust (DR) Estimator

Let $X_1, \dots, X_n$ be the observations of the aids data. Let θ be the parameters of the outcome and the exposure model, i.e., $\theta = (\alpha^T, \beta^T)$.

Let the parameter of interest be $ATE = g(\theta)$. The steps are as follows:

- ➊ **Draw** n observations $X_1^*, \dots, X_n^*$ with replacement from the data set.
- ➋ **Estimate coefficients:**
 - α for the exposure model $w(Z; \alpha)$.
 - β for the outcome model $\mu_a(Z; \beta)$, $a = 1, 2$.
- ➌ **Compute the DR estimator:** $g_b^*(\hat{\theta}(X_1^*, \dots, X_n^*))$.
- ➍ **Repeat steps 1–3:** B times, to get $g_1^*, \dots, g_B^*$.
- ➎ **Construct confidence intervals:**
 - Use empirical quantiles of $g_1^*, \dots, g_B^*$.
 - Or estimate the sample variance:

$$\frac{1}{B} \sum_{b=1}^B \left(g_b^* - \frac{1}{B} \sum_{b=1}^B g_b^* \right)^2.$$

```
# Load the data
aids1 = pd.read_csv("aids.txt", sep=" ")
# Set the random seed
random.seed(42)

B = 2000
bs_dr = np.empty(B)

for i in range(B):
# Generate bootstrap sample indices
ind = np.random.choice(a=len(aids1), size=len(aids1), replace=True)

# Use .iloc to index rows
aids = aids1.iloc[ind].copy()

# 1. Estimate the exposure model
# Logistic regression for exposure ( $X \sim Z$ )
X_exposure = add_constant(aids["Z"]) # Add intercept
y_exposure = aids["X"]
exp_m = Logit(y_exposure, X_exposure).fit(dis=0) # Silent fit
```

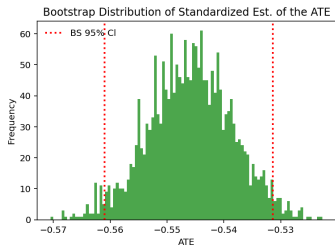
```
# Function to calculate propensity scores (w_Z)
def w_Z(Z):
    return expit(exp_m.params.iloc[0] + exp_m.params.iloc[1] * Z)

# 2. Estimate the outcome model
# Logistic regression for the outcome (Y ~ X + Z + X * Z)
aids.loc[:, "X_Z"] = aids["X"] * aids["Z"] # Interaction term
X_outcome = add_constant(aids[["X", "Z", "X_Z"]]) # Add intercept and terms
y_outcome = aids["Y"]
out = Logit(y_outcome, X_outcome).fit(dis=0) # Silent fit

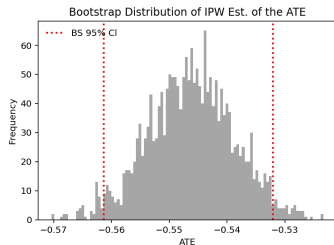
# Function to calculate predicted outcomes (out_X_Z)
def out_X_Z(X, Z):
    return expit(out.params.iloc[0] + out.params.iloc[1] * X + out.params.iloc[2] * Z
    + out.params.iloc[3] * X * Z)

# Compute the doubly robust average treatment effect (ATE)
bs_dr[i] = ( np.mean( aids["X"] * (aids["Y"]
- out_X_Z(1, aids["Z"])) / w_Z(aids["Z"])
+ out_X_Z(1, aids["Z"]) )
-
np.mean( (1 - aids["X"]) * (aids["Y"]
- out_X_Z(0, aids["Z"])) / (1 - w_Z(aids["Z"]))
+ out_X_Z(0, aids["Z"])) )
```

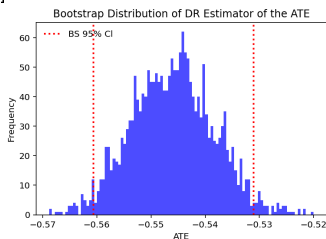
Bootstrapping the three estimators: Stand, IPW, DR



(a) 95% CI: [-0.561, -0.5314]



(b) 95% CI: [-0.5613, -0.5321]



(c) 95% CI: [-0.5606, -0.5311]

Due to its reliance on estimators from both outcome and exposure models, the DR estimator usually exhibits wider confidence intervals compared to non-robust estimators.

Summary of Doubly Robust Estimator

- The doubly robust estimator provides consistent estimates if:
 - The propensity score model is correctly specified, or
 - The outcome regression model is correctly specified.
- Improves efficiency compared to IPW alone, especially when the outcome model explains additional variation in Y .
- Practical recommendation:
 - Use doubly robust estimation as a safeguard against model misspecification.
 - Combine both models to leverage their strengths.
- The standard deviation, and consequently the confidence intervals (CIs), of the doubly robust (DR) estimator are generally larger than those of each individual non-robust estimator. This occurs because the DR estimator combines information from both the outcome model and the exposure model, which introduces additional variability compared to estimators that rely on only one of these models.

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- Standard methods for confounding control (e.g. stratification and outcome regression models) give conditional (subpopulation) causal effects
 - provided that we have conditional exchangeability, given the measured confounders
- Sometimes, the marginal (population) causal effect may be a more relevant target parameter
- Marginal causal effects can be estimated with standardization and IPW
 - either non-parametrically, or with regression models

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